

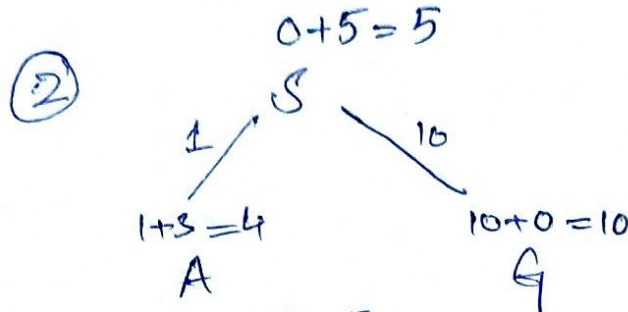
Artificial Intelligence

Applied Artificial Intelligence

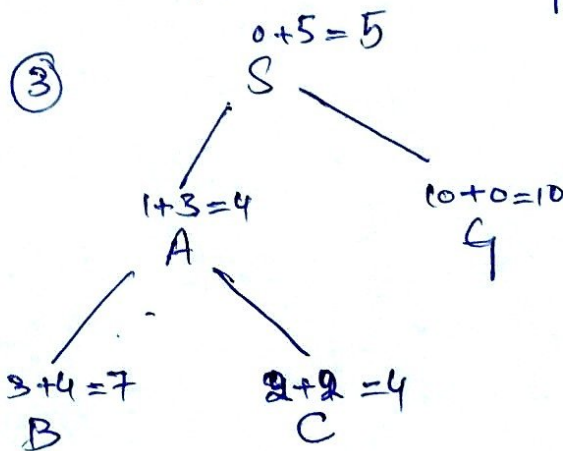
Sessional - II Solution

Question 1:- ①

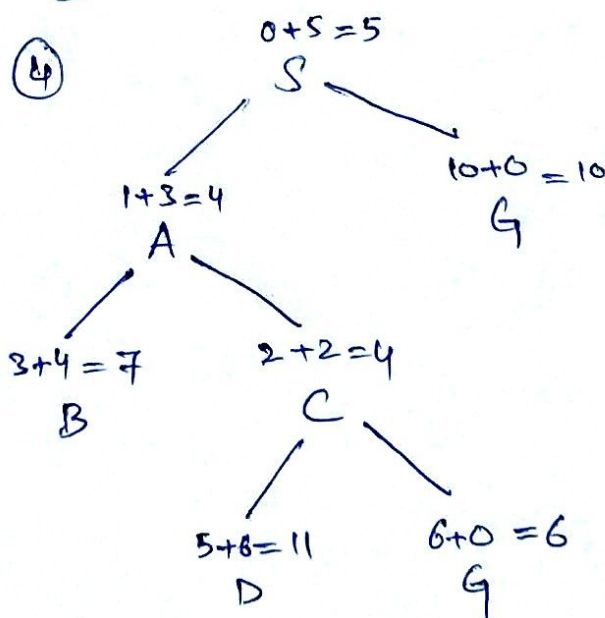
Fringe



Fringe
 $S \rightarrow A : 4$
 $S \rightarrow G : 10$



Fringe
 $S \rightarrow G : 10$
 $S \rightarrow A \rightarrow B : 7$
 $S \rightarrow A \rightarrow C : 4$



Fringe
 $S \rightarrow G : 10$
 $S \rightarrow A \rightarrow B : 7$
 $S \rightarrow A \rightarrow C \rightarrow D : 11$
 $S \rightarrow A \rightarrow C \rightarrow G : 6$

G will be ~~expand~~ selected from fringe.

As G is the goal

$\Rightarrow S \rightarrow A \rightarrow C \rightarrow G$ is the Chosen path with the cost = 6.

Question 2 :- For loan approved $P(\text{Yes}) = 5/8$
 $P(\text{No}) = 3/8$

<u>Income level</u>		
	$P(\text{Yes})$	$P(\text{No})$
High	$3/5$	$0/3$
Low	$1/5$	$2/3$
Medium	$1/5$	$1/3$

<u>Credit Score</u>		
	$P(\text{Yes})$	$P(\text{No})$
Excellent	$3/5$	$0/3$
Poor	$0/5$	$2/3$
Fair	$2/5$	$1/3$

<u>Loan Purpose</u>		
	$P(\text{Yes})$	$P(\text{No})$
Business	$2/5$	$1/3$
Personal	$2/5$	$1/3$
Education	$1/5$	$1/3$

<u>Previous Default</u>		
	$P(\text{Yes})$	$P(\text{No})$
Yes	$0/5$	$3/3$
No	$5/5$	$0/3$

$X = \begin{matrix} \text{Income} \\ \downarrow \\ \text{Medium} \end{matrix}, \begin{matrix} \text{Credit} \\ \downarrow \\ \text{Fair} \end{matrix}, \begin{matrix} \text{Loan purpose} \\ \downarrow \\ \text{Education} \end{matrix}, \begin{matrix} \text{Prev. Default} \\ \downarrow \\ \text{No} \end{matrix}$

$$P(\text{Yes}|X) = \frac{1}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{5}{5} \times \frac{5}{8}$$

$$= 0.2 \times 0.4 \times 0.2 \times 1 \times 0.625$$

$$P(\text{Yes}|X) = 0.01$$

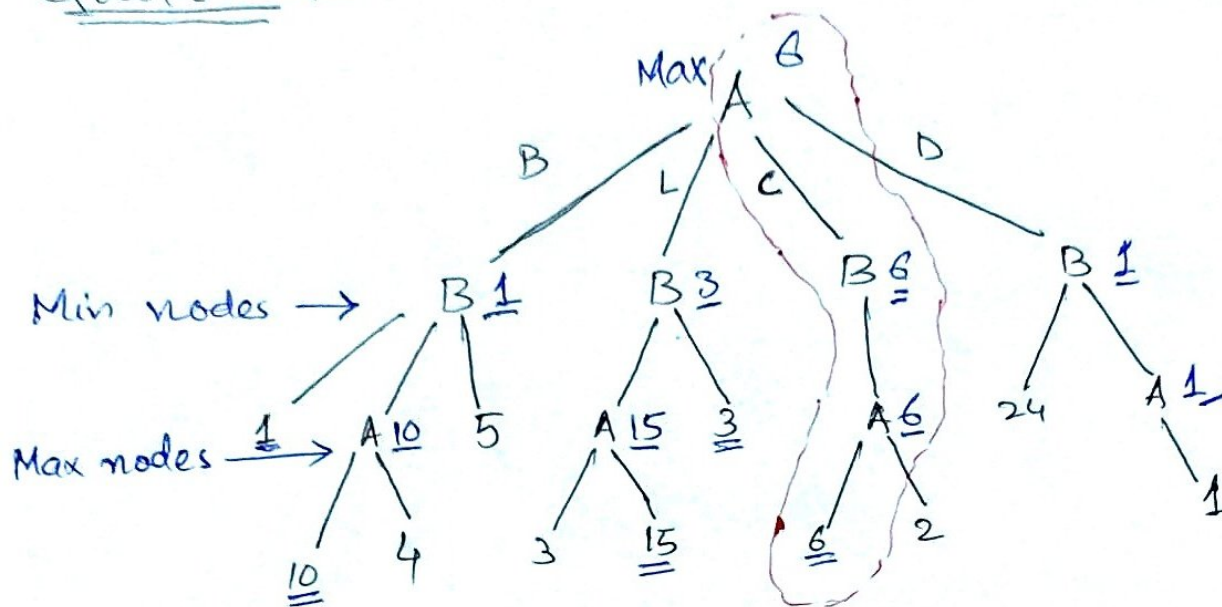
$$P(\text{No}|X) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times 0 \times \frac{3}{8}$$

$$P(\text{No}|X) = 0$$

$$P(\text{Yes}|X) > P(\text{No}|X)$$

$\Rightarrow$ Loan should be approved!

Question 3:-



⇒ Zalmi, Compete against all odds.